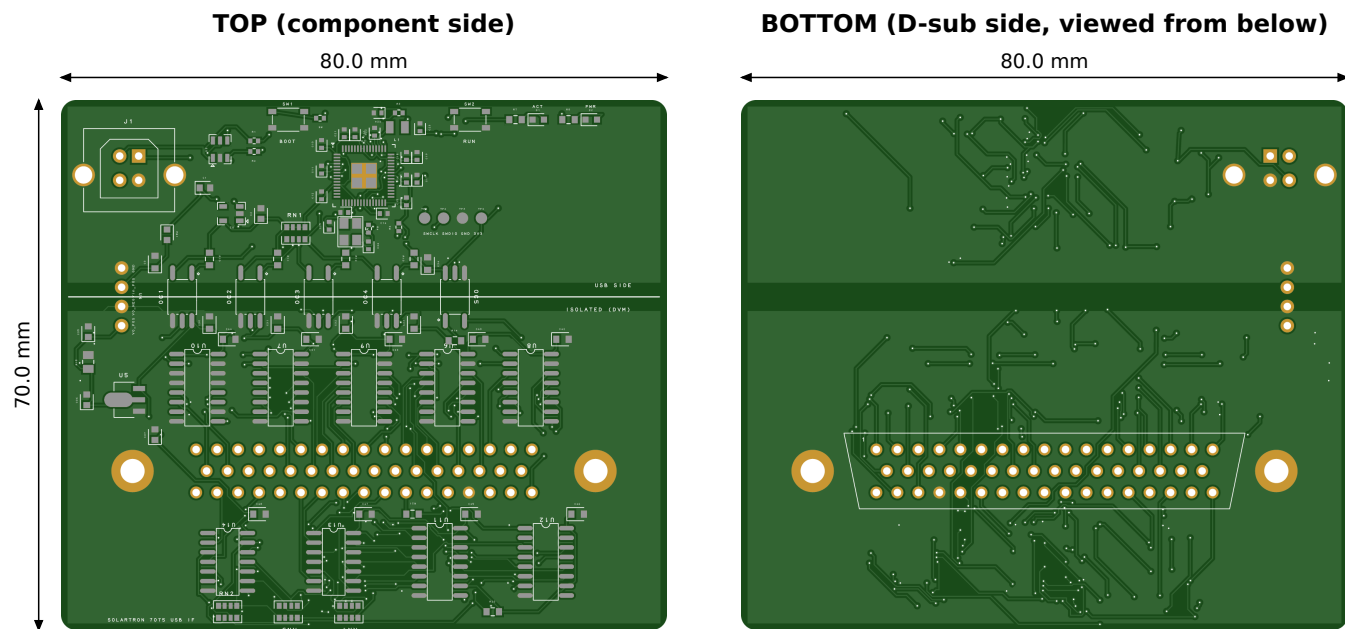


Check the print: this bar must measure 50 mm and the board outlines 80 x 70 mm.
Cut out the bottom view to check the DD-50 against the 70754's SKB socket.

Board views

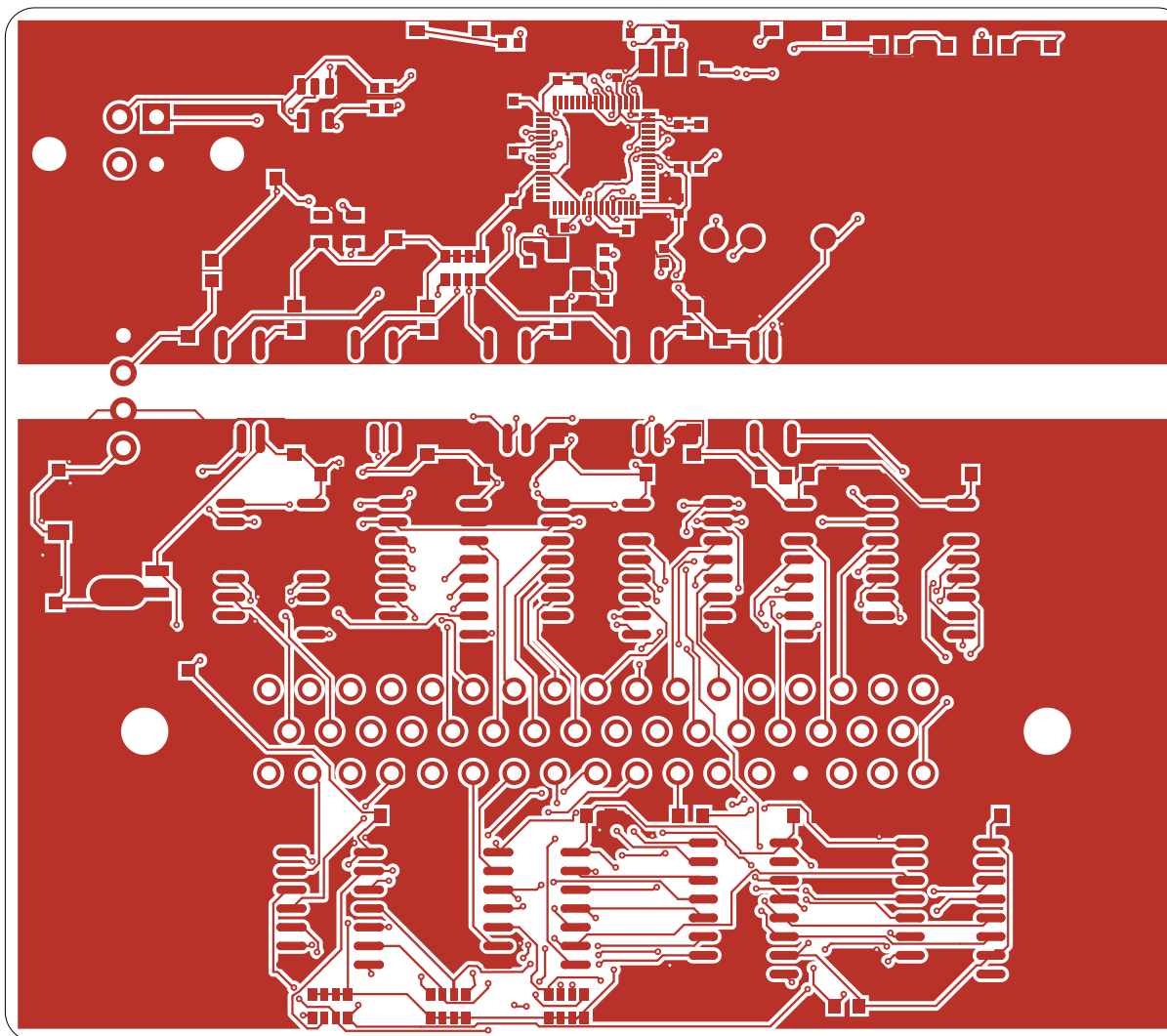
Rendered from the Gerbers with
pcb-stackup (tracespace).

Top: all SMD parts, USB-B (J1),
B0505S-1WR3 (U4).

Bottom: DD-50 plug (J2) only.

Isolation barrier: the silkscreen
line across the optocouplers.
USB side above, DVM side below.

Project	Solartron 7075 USB interface (RP2354A)
Sheet	Board views, actual size
View	top from above; bottom from below (mirrored)
Scale	1:1
Board	80 x 70 mm, 2 layers, 1.6 mm FR-4
Source	tscircuit/outputs/gerbers.zip
Print at 100 % (actual size)	



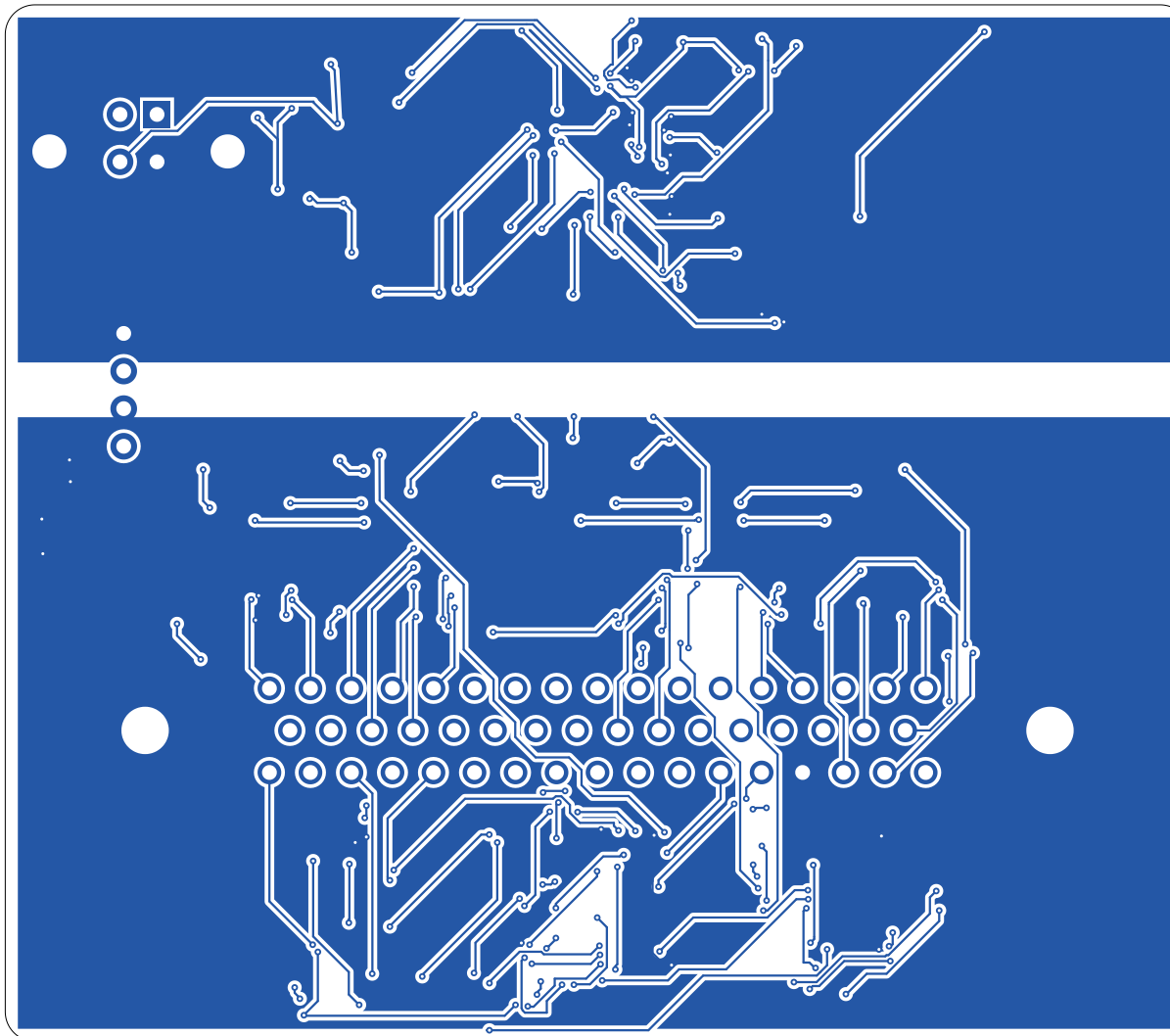
0 5 10 15 20 mm

Top copper

Tracks 0.15 mm min, vias
0.2 mm drill / 0.45 mm pad.
Pours: GND (USB side) and
GND_ISO (DVM side), stopping
1.6 mm each side of the barrier.

Drilled holes shown white.

Project	Solartron 7075 USB interface (RP2354A)
Sheet	Top copper (F.Cu)
View	from top
Scale	2:1
Board	80 x 70 mm, 2 layers, 1.6 mm FR-4
Source	tscircuit/outputs/gerbers.zip
Print at 100 % (actual size)	



Bottom copper

Tracks 0.15 mm min, vias
0.2 mm drill / 0.45 mm pad.
Pours: GND (USB side) and
GND_ISO (DVM side), stopping
1.6 mm each side of the barrier.

Drilled holes shown white.

Project	Solartron 7075 USB interface (RP2354A)
Sheet	Bottom copper (B.Cu)
View	from top (seen through the board)
Scale	2:1
Board	80 x 70 mm, 2 layers, 1.6 mm FR-4
Source	tscircuit/outputs/gerbers.zip
Print at 100 % (actual size)	

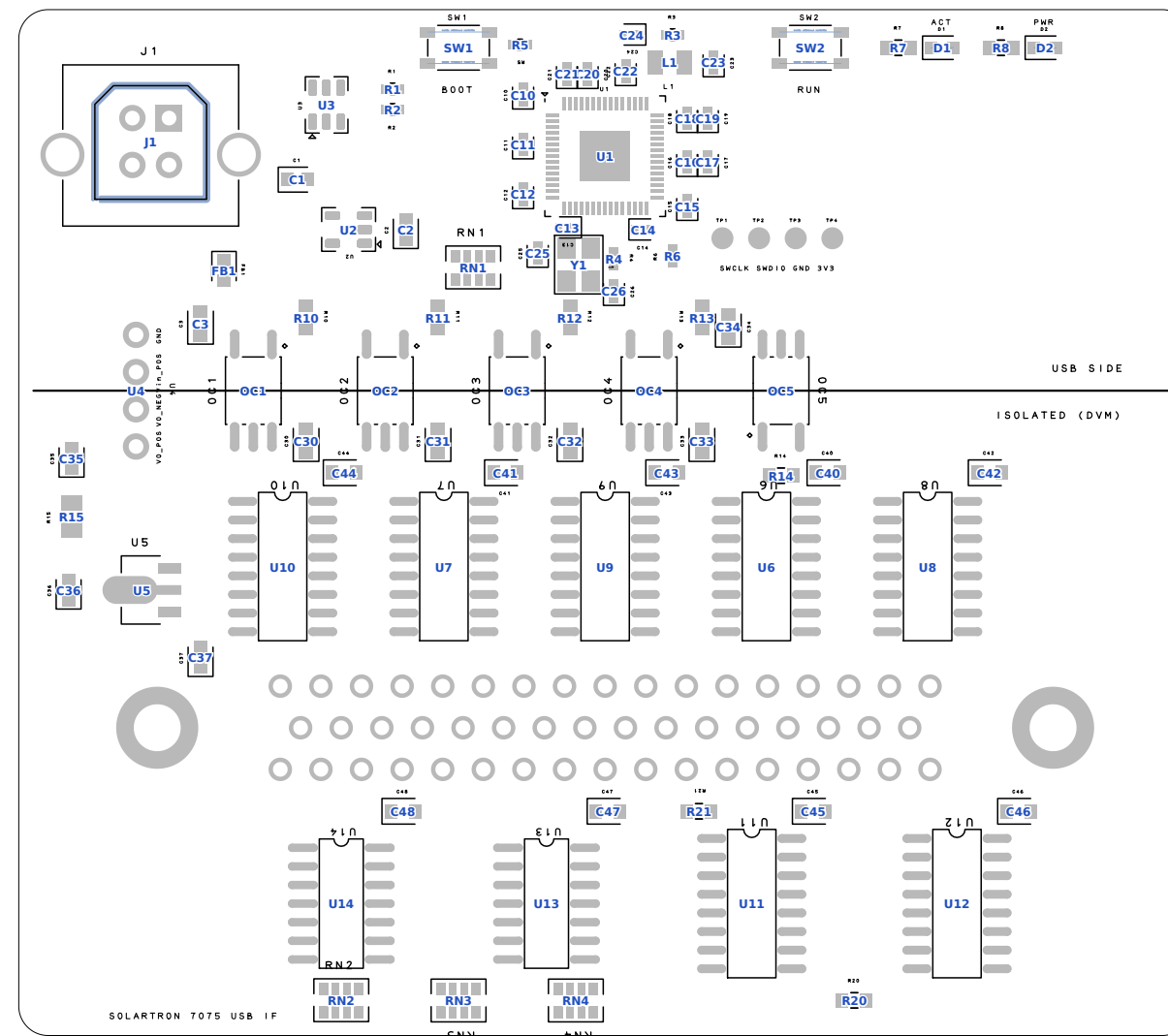
Top assembly

All SMD parts are on this side.

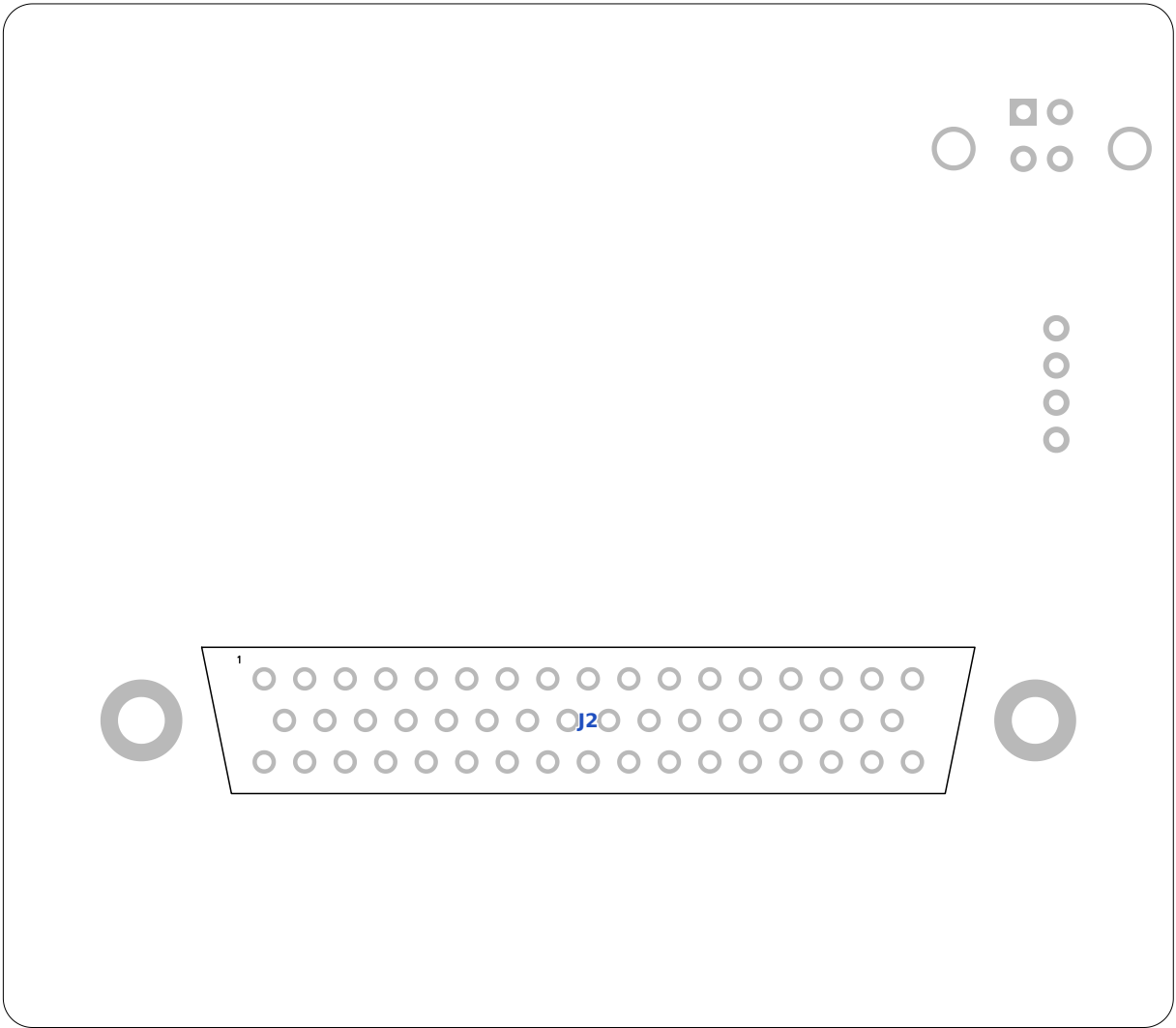
J1 (USB-B) and U4 (B0505S) are through-hole, fitted from this side.

Designators in blue are at the part centres in pick_and_place.csv.

- Reference designator
- Silkscreen
- Pads (solder-mask openings)
- Fabrication outlines (F.Fab)



Project	Solartron 7075 USB interface (RP2354A)
Sheet	Top assembly
View	from top
Scale	2:1
Board	80 x 70 mm, 2 layers, 1.6 mm FR-4
Source	tscircuit/outputs/gerbers.zip

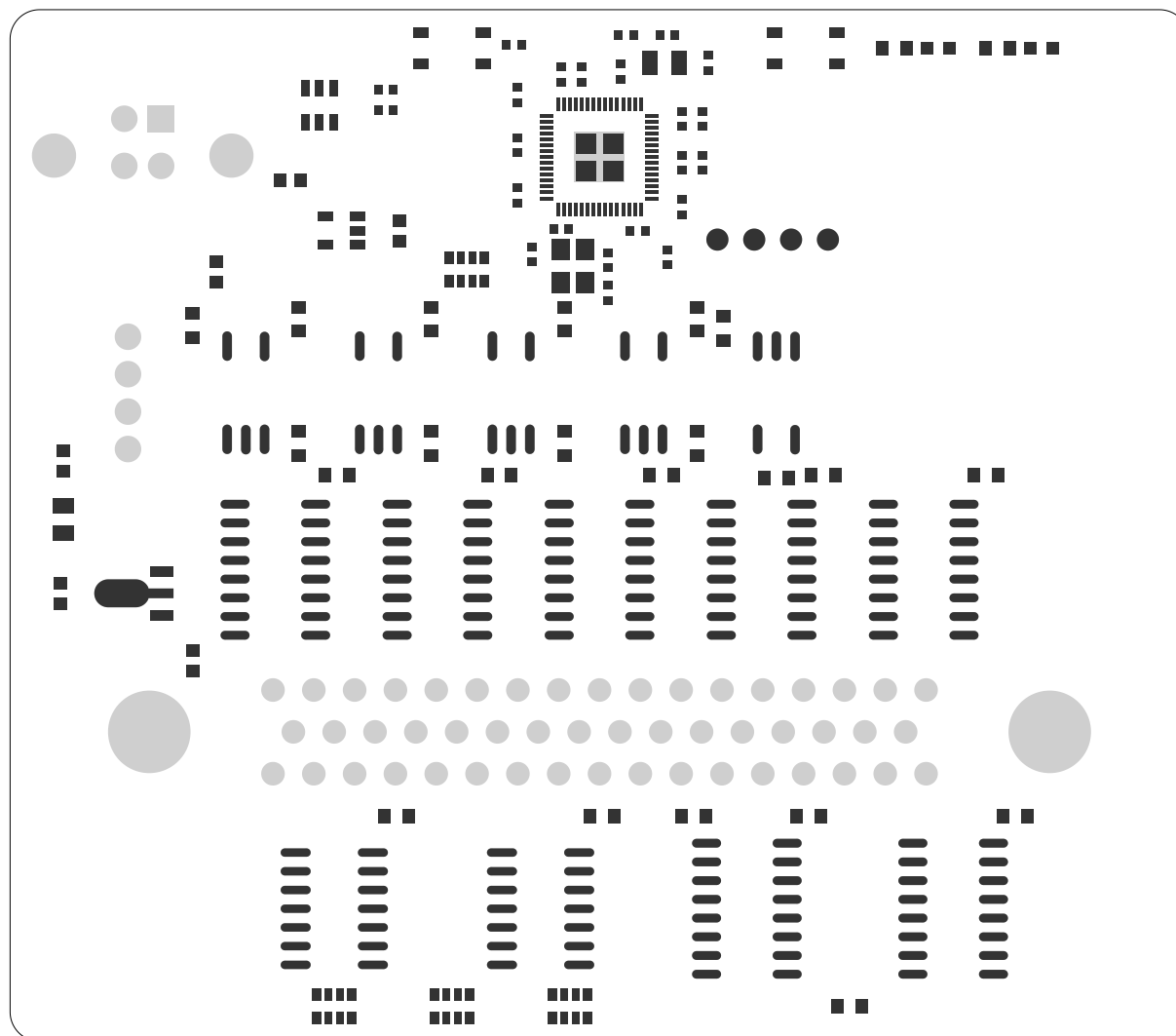


Bottom assembly

J2 (DD-50 plug) is the only part on this side; solder it from the top. Trim J1/U4 leads flush: this face sits a few mm from the 70754 housing.

- Reference designator
- Silkscreen
- Pads (solder-mask openings)

Project	Solartron 7075 USB interface (RP2354A)
Sheet	Bottom assembly
View	from below (mirrored)
Scale	2:1
Board	80 x 70 mm, 2 layers, 1.6 mm FR-4
Source	tscircuit/outputs/gerbers.zip



Top stencil

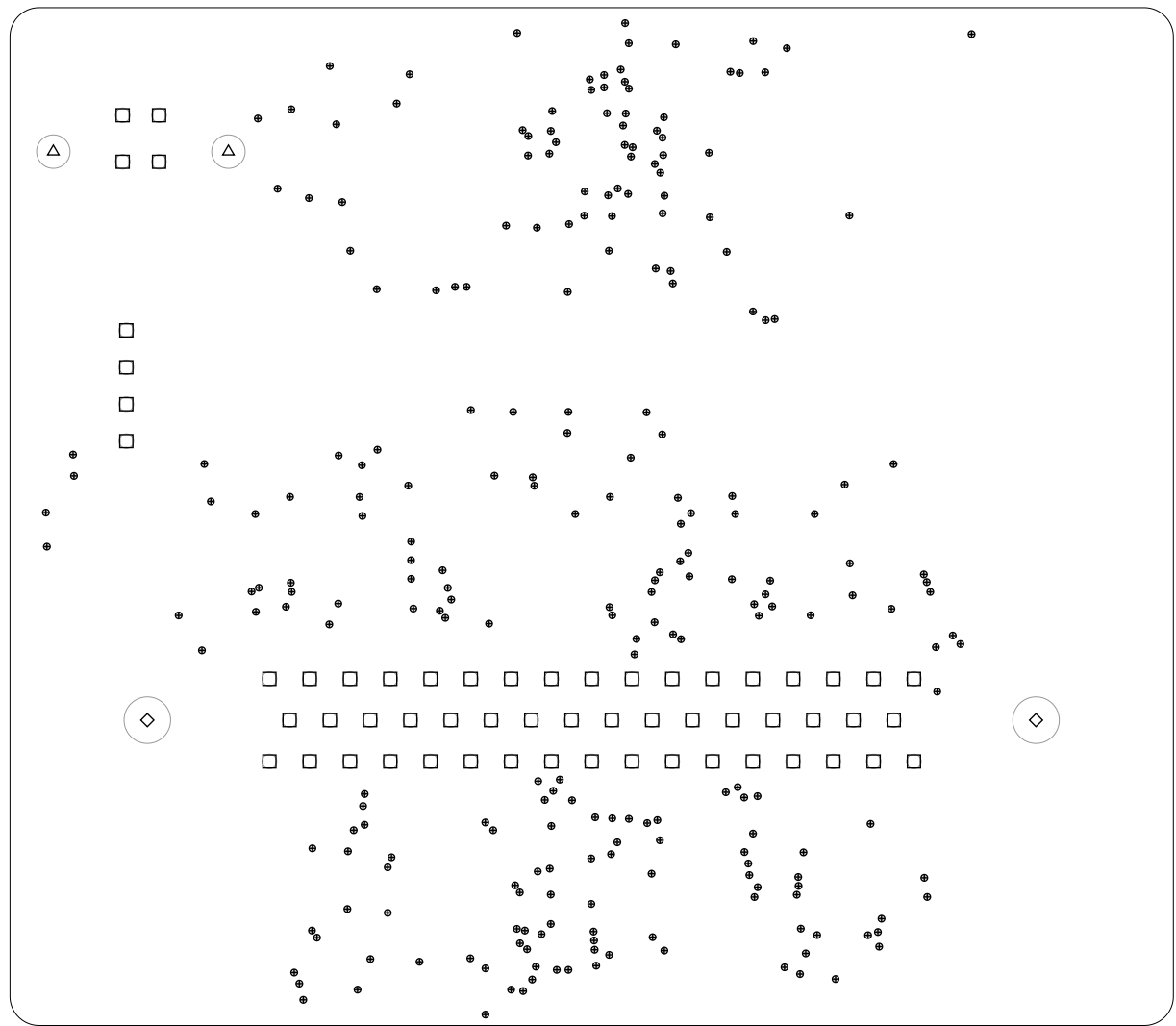
Apertures 1:1 with the pads;
 RP2354A exposed pad: 2 x 2
 windowpane, about 60 %.
 No paste on through-holes.
 No bottom stencil needed.

Made by scripts/prepare-fab.mjs
 (tscircuit omits paste on pill
 pads: every SOIC and TLP2361).

- Paste apertures
- Pads (mask openings)

0 5 10 15 20 mm

Project	Solartron 7075 USB interface (RP2354A)
Sheet	Top stencil (F.Paste)
View	from top
Scale	2:1
Board	80 x 70 mm, 2 layers, 1.6 mm FR-4
Source	tscircuit/outputs/gerbers.zip
Print at 100 % (actual size)	



Drill drawing

Sym	Finished dia.	Qty	Plating	Use
⊕	0.20 mm	248	PTH	vias
□	1.00 mm	58	PTH	J1, J2, U4 pins
△	2.30 mm	2	PTH	J1 shell
◇	3.20 mm	2	PTH	J2 flange (4-40)

Total 310

All holes plated through.
Sizes are finished hole diameters.
File: drill-L1-L2.drl (Excellon, mm).

Project	Solartron 7075 USB interface (RP2354A)
Sheet	Drill drawing
View	from top
Scale	2:1
Board	80 x 70 mm, 2 layers, 1.6 mm FR-4
Source	tscircuit/outputs/gerbers.zip